

☐ (a)
Interference

DMax Hole = 7,015 mm

Dmin Hole = 7,000 mm

dMax Shaft = 6,995 mm

dMin Shaft = 6,986 mm

☐ (b)
Clearance

DMax Hole = 7,009 mm

Dmin Hole = 7,000 mm

dMax Shaft = 7,995 mm

dMin Shaft = 7,989 mm

☐ (c)

Dmin Hole = 7,000 mm

dMax Shaft = 6,995 mm

dMin Shaft = 6,986 mm

☐ (b)

Clearance

DMax Hole = 7,009 mm

Dmin Hole = 7,000 mm

dMax Shaft = 7,995 mm

dMin Shaft = 7,989 mm

☐ (c)

Clearance

DMax Hole = 7,015 mm

Dmin Hole = 7,000 mm

dMax Shaft = 6,995 mm

dMin Shaft = 6,986 mm

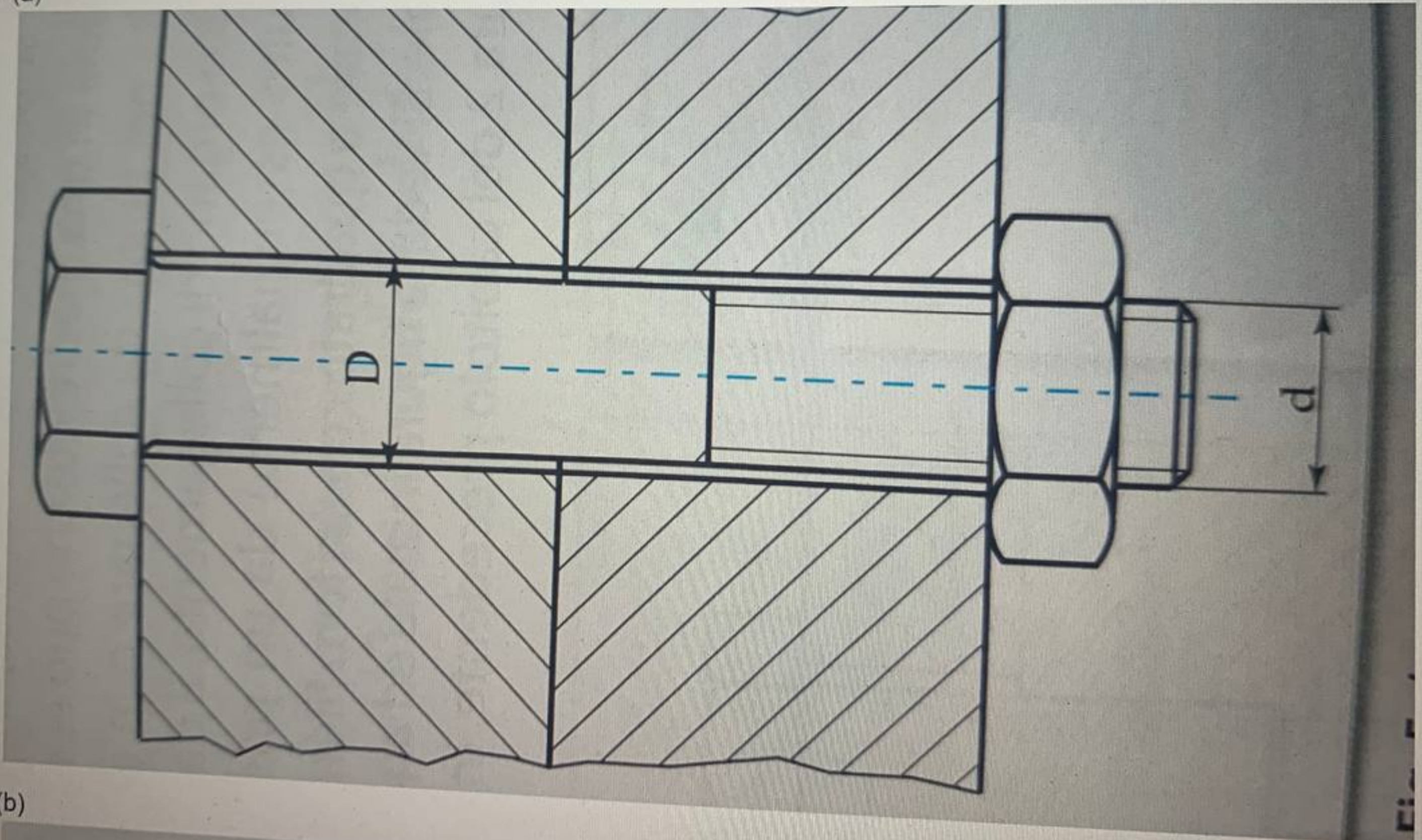
Engineering Drawing

What is the meaning of the dimension: G 1 ?

- ☐ (a)
Not Sealed GAS Threadings with nominal diameter of 1 inch
- ☐ (b)
ISO Threadings with nominal diameter of 1 mm
- ☐ (c)
Sealed GAS Threadings with nominal diameter of 1 inch

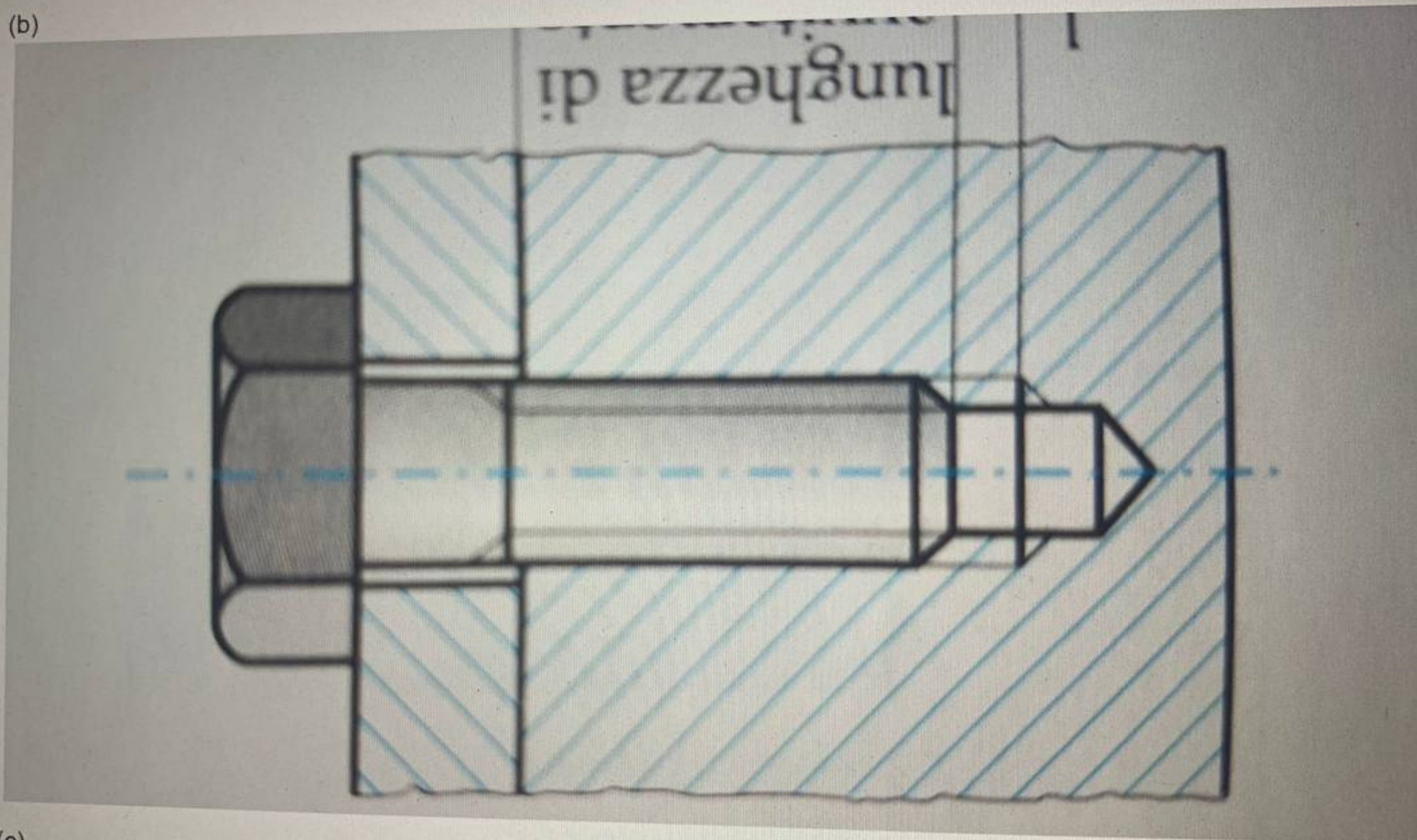
Which of the following drawings represents a mechanical connection with a cap screw

☐ (a)



☐ (b)

○ (b)

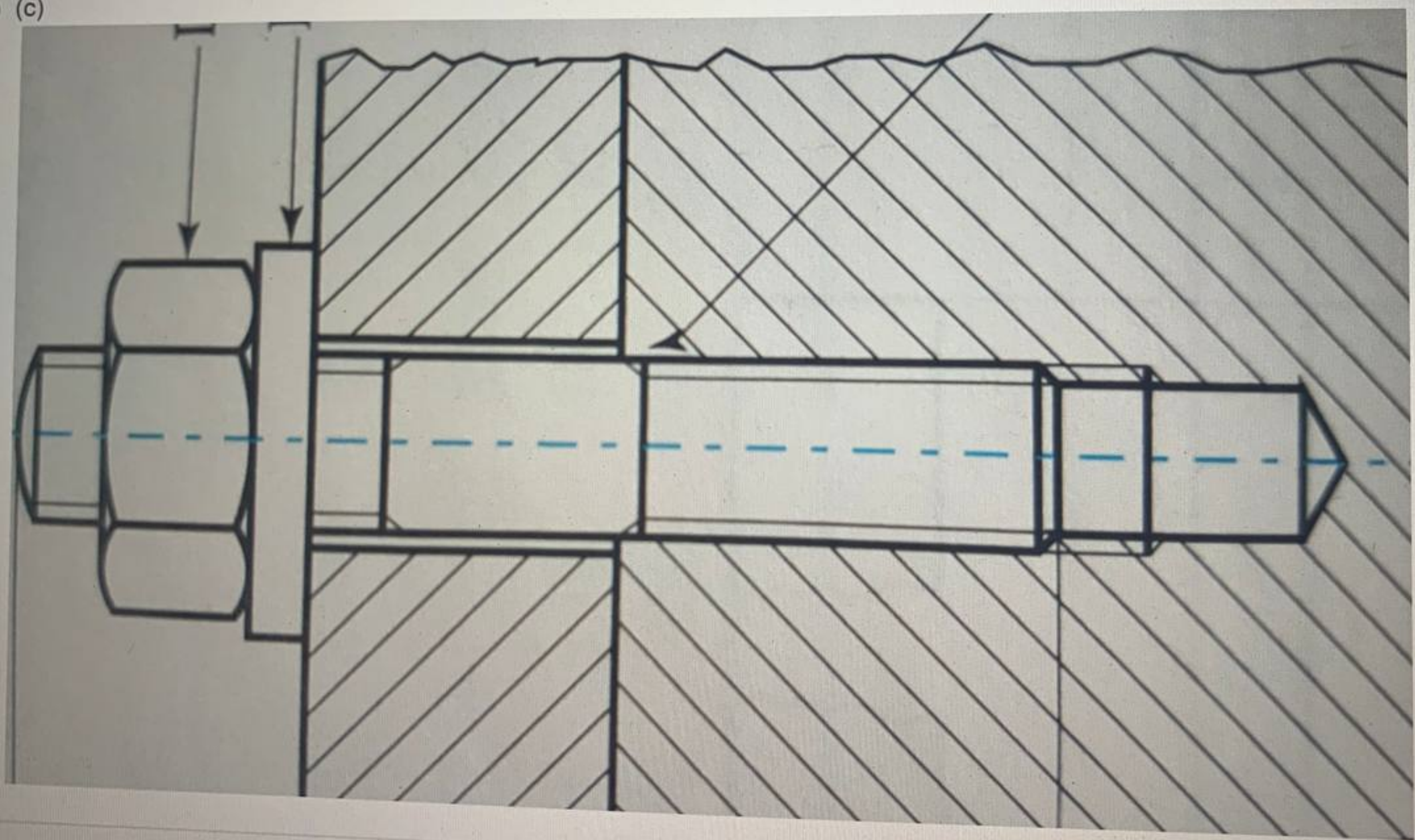


○ (c)



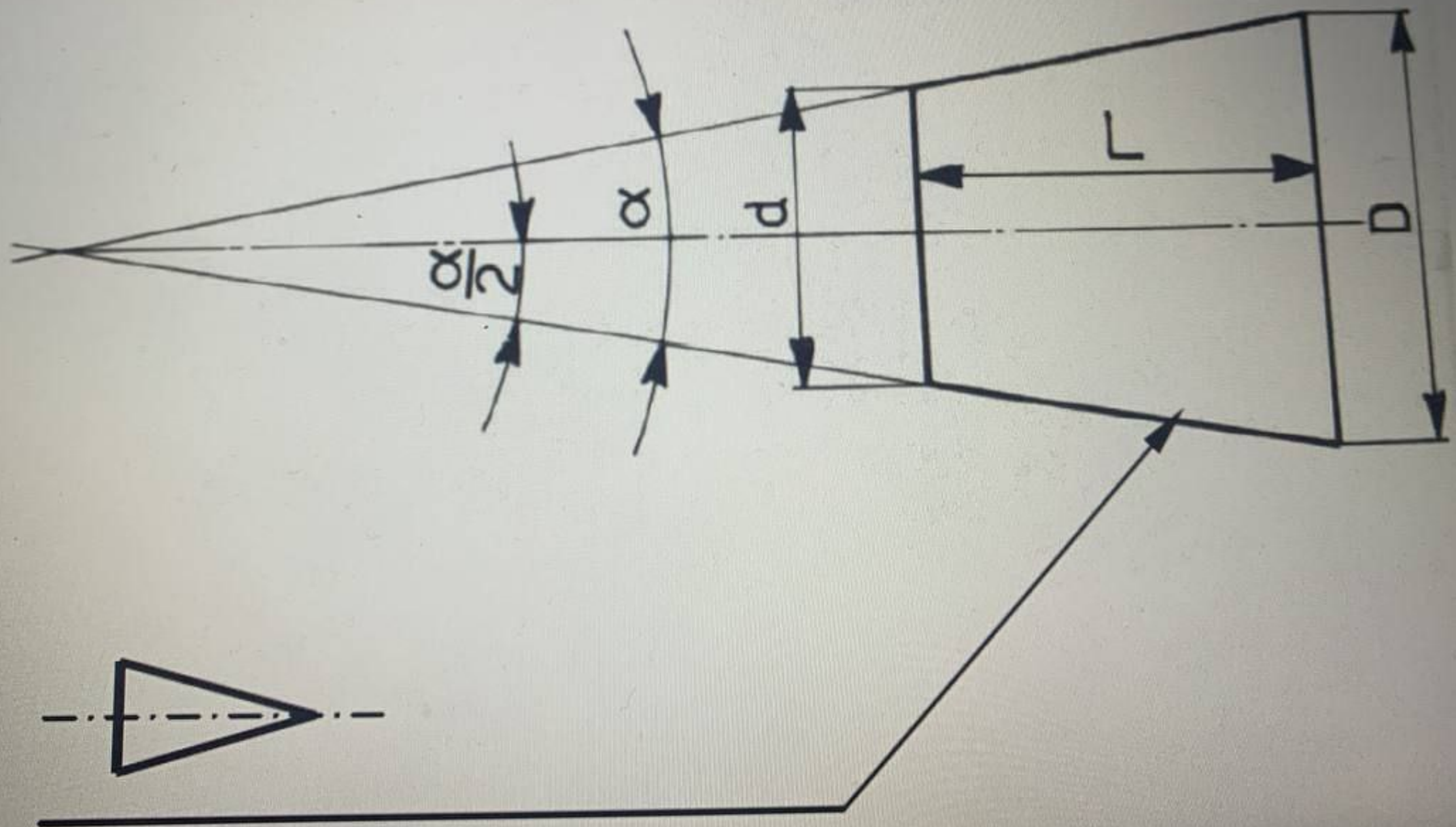


○ (c)



Engineering Drawing

Given the 1:4 taper, given the length $L = 10$, given $D = 60$ calculate the smaller diameter $d =$



$$f. \quad \epsilon \frac{\alpha}{2} = \frac{D-d}{2L}$$

$$E_s = 1\bar{1} + E_i$$

$$E_i = E_s - 1\bar{1}$$

$$P_i = E_s - 1\bar{1}$$

1. TIO 2010/1

2. v. 5000

3. Butness

C/P.

$$E_s - P_i$$

inches

$$1 E_i - P_i$$

R = Gas

w = axis

inches

inches

coprocessor.

BO1A

capable screen.

4. 5 years

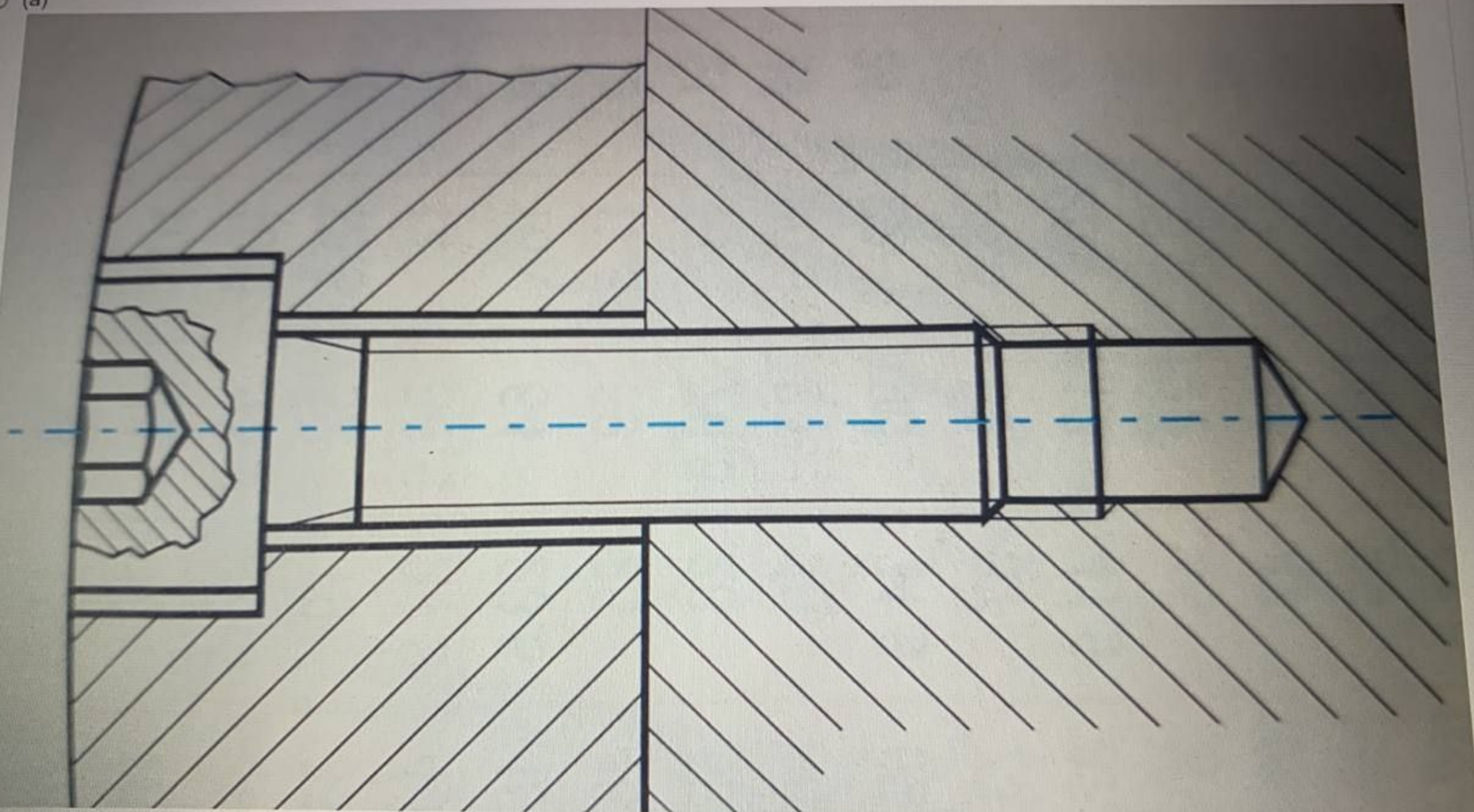
5. KNAUF

$$D_{max} = D + E_i$$

$$D_{min} = D + E_i$$

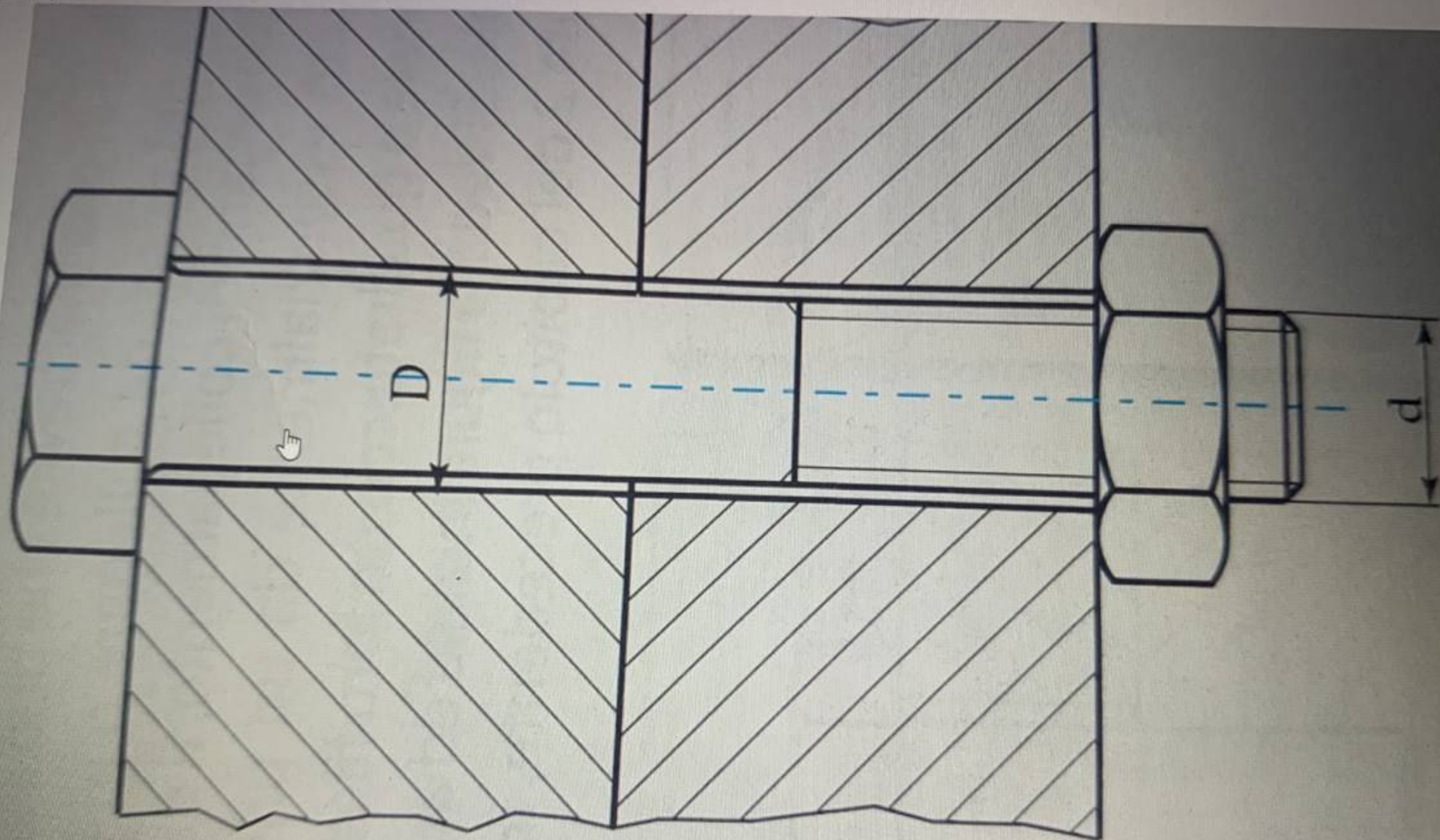
Which of the following drawings represents a mechanical component?
Cylindrical head with hexagonal slot

☐ (a)



(b)

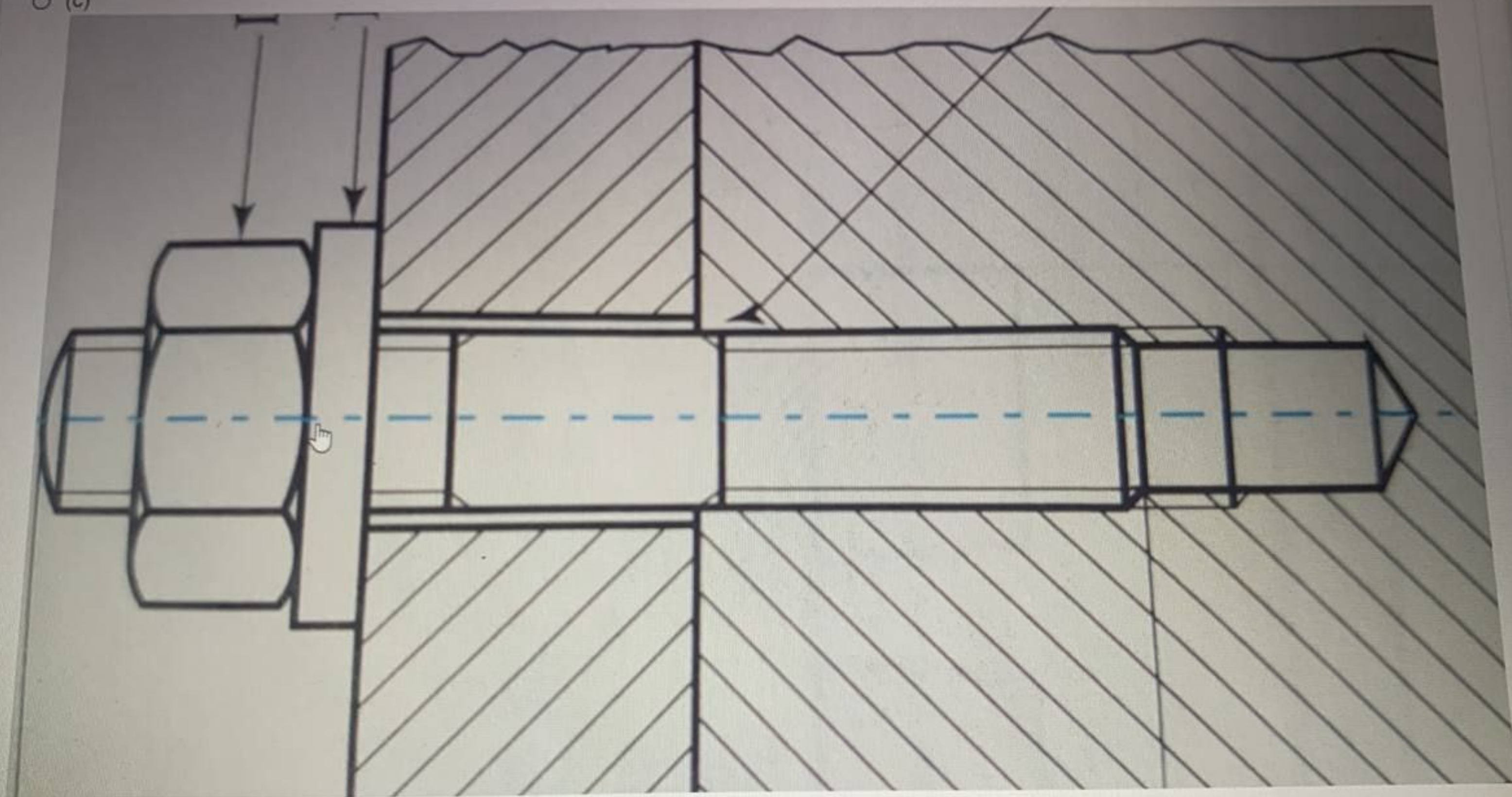
○ (b)



○ (c)



○ (c)



Domanda 4

Risposta non
ancora dataPunteggio max.:
5,00Contrassegna
domanda

What is the meaning of the dimension: 10H7 ?

☐ (a)

Dimensional Tolerance with position H and grade 7 for a dimension 10 mm

☐ (b)

ISO Threadings with nominal diameter 10 and fine pitch 7 mm

☐ (c)

Withworth Threading with nominal diameter 10 inch

Precedente

Successivo

What is the meaning of the dimension: R 1/2 ?

- ☐ (a)
Sealed GAS Threading with nominal diameter 1/2 inch
- ☐ (b)
Dimensional Tolerance with position R and grade 1 for a dimension 2
- ☐ (c)
ISO Threading with nominal diameter 1 mm and pitch 2 mm

Domanda 3

Risposta non
ancora data

Punteggio max.:
5,00

Contrassegna
domanda

Indicates and explains the different Threaded fasteners definitions



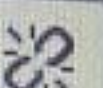
Indicates and explains which are the main dimensions of internal and external threads and which are the common thread forms



A ▼

B

I



Employing the following table, that explain holes and shaft deviations and tolerance grades calculate the following fit: $\phi 12 H6 / s5$, specifying if it is interference or clearance, indicating the maximum and minimum diameter of the hole and shaft

Basic size mm	Upper deviations e_s													Lower deviations e_i												
	Tolerance grades (all)													Tolerance grades (all)												
Above	Up to	h ¹⁾	b ¹⁾	c	cd	d	e	ef	f	fg	g	h	js ²⁾	j	k	m	n	p	r	s	t	u	v	x	y	z
-	3 ¹⁾	-270	-140	-60	-34	-20	-14	-10	-6	-4	-2	0		-2	-4	-6	0	0	+2	+4	+6	+10	+14		+18	+20
3	6	-270	-140	-70	-46	-30	-20	-14	-10	-6	-4	0		-2	-4	+1	0	0	+4	+8	+12	+15	+19		+23	+26
6	10	-280	-150	-80	-56	-40	-25	-18	-13	-8	-5	0		-2	-5	+1	0	0	+6	+10	+15	+19	+23		+28	+35
10	14																									
14	18	-290	-150	-95		-50	-32		-16		-6	0		-3	-6	+1	0	0	+7	+12	+18	+23	+28		+33	+40
																										+45
																										+60

Dimensional class		Tolerance grade										
mm		IT1	IT2	IT3	IT4	IT5	IT6	IT7	IT8	IT9	IT10	IT11
Above	Up to	Tolerance										
	μm											
-	3	0.8	1.2	2	3	4	6	10	14	25	40	60
3	6	1	1.5	2.5	4	5	8	12	18	30	48	75
6	10	1	1.5	2.5	4	6	9	15	22	36	58	90
10	18	1.2	2	3	5	8	11	18	27	43	70	110

Basic size		Lower deviations E_i										
mm		Tolerance grades (all)										
Above	Till to	A ¹⁾	B ¹⁾	C	CD	D	E	EF	F	FG	G	H
-	3	+270	+140	+60	+34	+20	+14	+10	+6	+4	+2	0
3	6	+270	+140	+70	+46	+30	+20	+14	+10	+6	+4	0
6	10	+280	+150	+80	+56	+40	+25	+18	+13	+8	+5	0
10	14											
14	18	+290	+150	+95		+50	+32		+18		+6	0

☐ (a)
Clearance

DMax Hole = 12,011 mm

Dmin Hole = 12,000 mm

dMax Shaft = 12,036 mm

dMin Shaft = 12,028 mm

☐ (b)
Interference

DMax Hole = 12,011 mm

Dmin Hole = 12,000 mm

dMax Shaft = 12,036 mm

dMin Shaft = 12,028 mm

☐ (b)
Interference

DMax Hole = 12,011 mm

Dmin Hole = 12,000 mm

dMax Shaft = 12,036 mm

dMin Shaft = 12,028 mm

☐ (c)

Interference

DMax Hole = 12,009 mm

Dmin Hole = 12,000 mm

dMax Shaft = 12,021 mm

dMin Shaft = 12,015 mm

Precedente

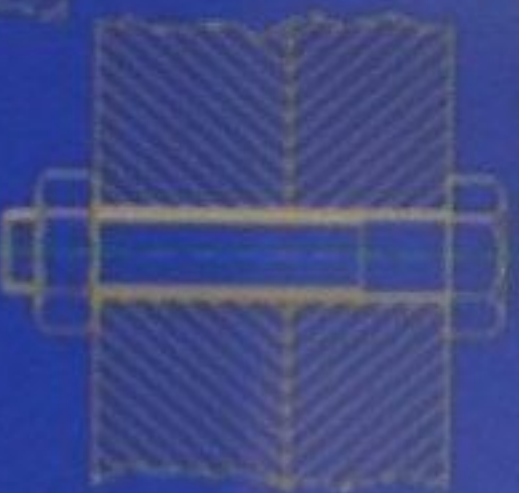
Successivo

Fastener definitions



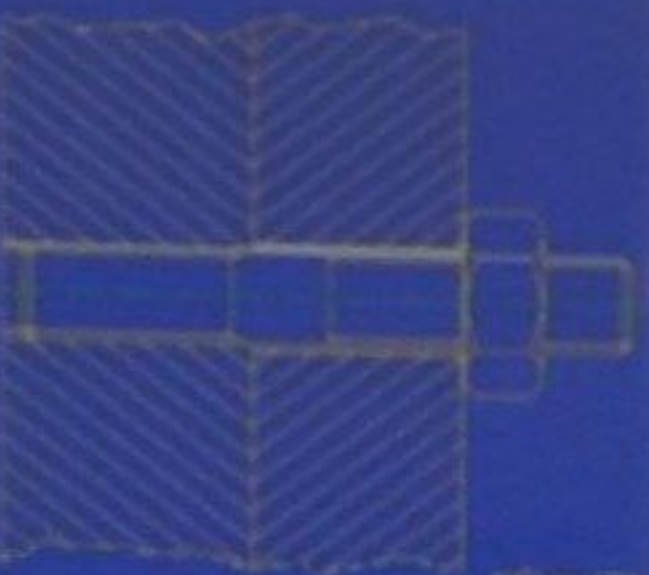
(It is a threaded fastener that joins two or more parts by passing through a clearance hole in one part and screwing into a threaded hole in the other)

Cap screw



(A bolt is a threaded fastener that passes through clearance holes in assembled parts and threads into a nut)

Bolt

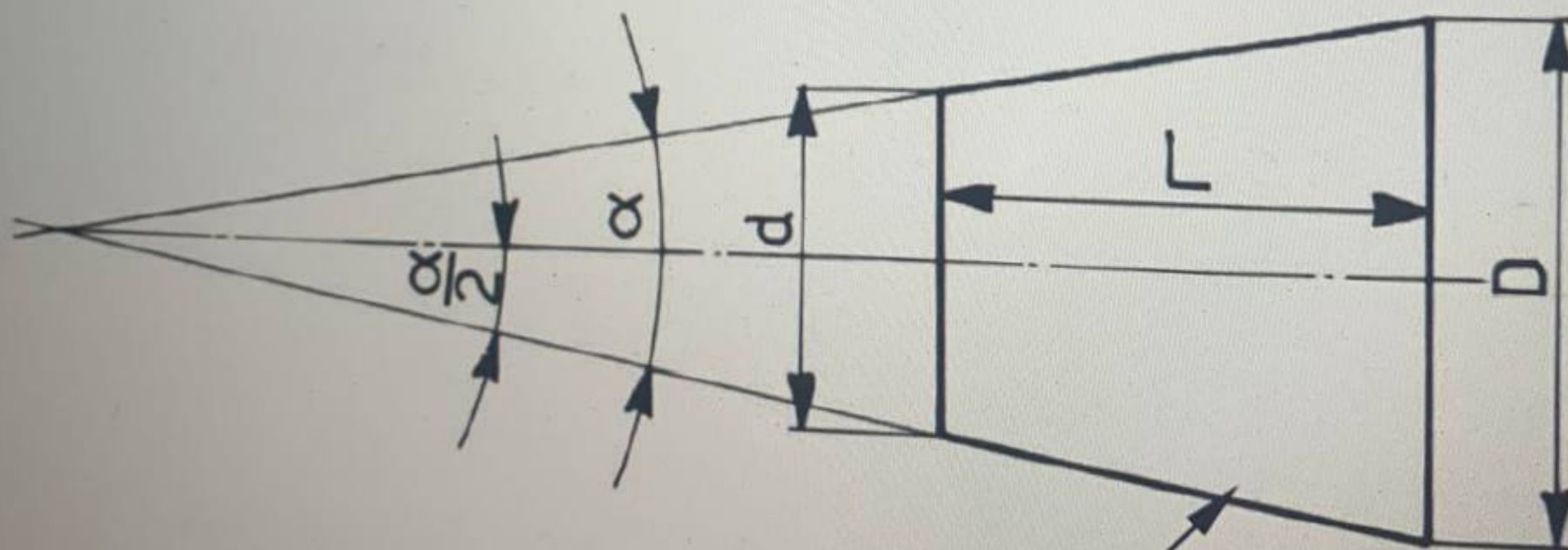


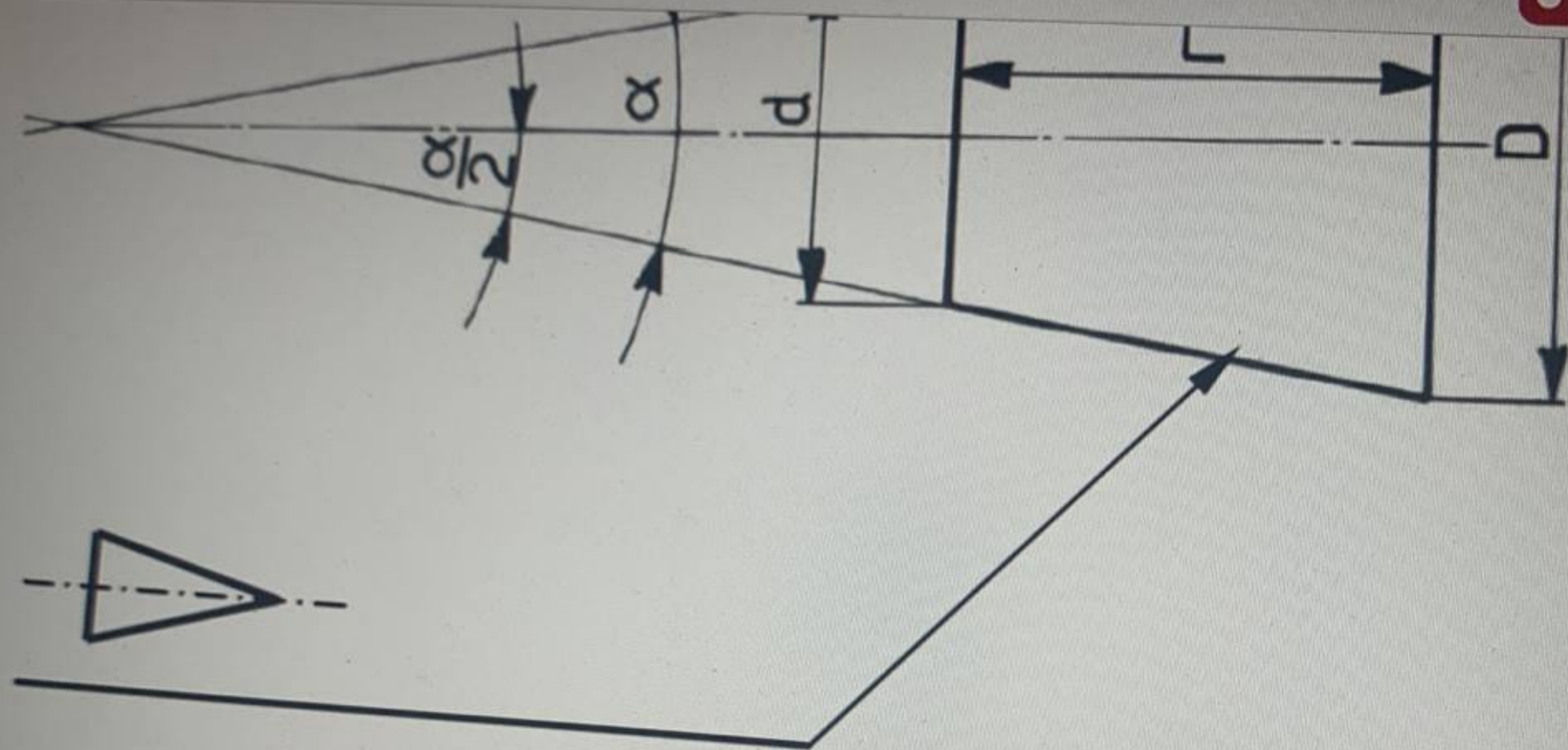
Captive screw

(Are those that remain attached to one part even when disengaged from the other mating part)

Engineering Drawing

Given the 1:4 taper, given the length $L = 100$, given $D = 80$ calculate the smaller diameter $d =$





Scegli una o più alternative:

- ☐ (a) 55
- ☐ (b) 25
- ☐ (c) 32
- ☐ (d) 43,75

②. $\phi 7 H7/g6$

Clearance

$$\begin{aligned} D_{\text{Max Hole}} &= 7.015 \\ D_{\text{Min Hole}} &= 7.000 \\ d_{\text{Max Shaft}} &= 6.995 \\ d_{\text{Min Shaft}} &= 6.986 \text{ mm} \end{aligned}$$

③. $\phi 11 H6/g6$

Clearance

$$\begin{aligned} D_{\text{Max Hole}} &= 11.011 \\ D_{\text{Min Hole}} &= 11.000 \\ d_{\text{Max Shaft}} &= 10.994 \\ d_{\text{Min Shaft}} &= 10.983 \end{aligned}$$

④. $\phi 12 H6/s5$

Interference

$$\begin{aligned} D_{\text{Max Hole}} &= 12.011 \\ D_{\text{Min Hole}} &= 12.000 \\ d_{\text{Max Shaft}} &= 12.036 \\ d_{\text{Min Shaft}} &= 12.028 \end{aligned}$$

⑥. $D = d + \frac{L}{k}$

$$\begin{array}{l|l} L=10 & \\ D=60 & \\ k=4 & \\ \hline d=? & \end{array}$$

Taper: 1:4.

$$60 = d + \frac{10}{4}$$

$$60 = d + 2.5$$

$$d = 57.5$$

$$\boxed{d = 57.5}$$

⑤. Dimension: 10 H7

Dimensional Tolerance with position H and grade 7 for a dimension 10 mm.

Employing the following table, that explain holes and shaft deviations and tolerance grades calculate the following fit: $\phi 7$ H7 / g6, specifying if it is interference or clearance, indicating the maximum and minimum diameter of the hole and shaft

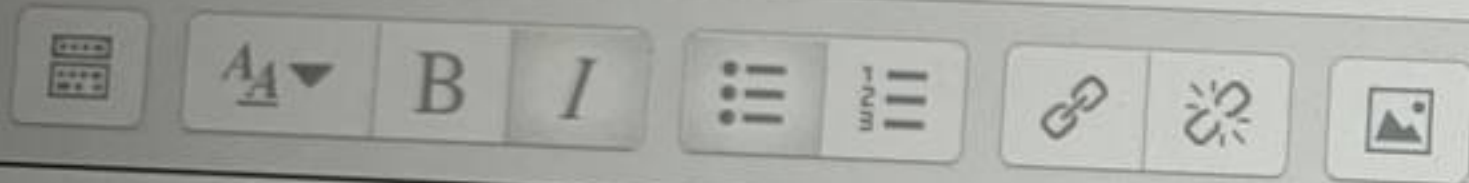
Basic size mm		Upper deviations <i>es</i>												Lower deviations <i>ei</i>																
		Tolerance grades (all)												IT5 e IT6	IT7	IT8	IT4 a IT7	fino a IT3 e sopra a IT7	Tolerance grades (all)											
Above	Up to	a ¹⁾	b ¹⁾	c	cd	d	e	el	f	fg	g	h	js ²⁾	j			k		m	n	p	r	s	t	u	v	x	y	z	
-	3 ¹⁾	-270	-140	-60	-34	-20	-14	-10	-6	-4	-2	0		-2	-4	-6	0	0	+2	+4	+6	+10	+14		+18		+20		+26	
3	6	-270	-140	-70	-46	-30	-20	-14	-10	-6	-4	0		-2	-4		+1	0	+4	+8	+12	+15	+19		+23		+28		+35	
6	10	-280	-150	-80	-56	-40	-25	-18	-13	-8	-5	0		-2	-5		+1	0	+6	+10	+15	+19	+23		+28		+34		+42	
10	14																													
14	18	-290	-150	-95		-50	-32		-16		-6	0		-3	-6		+1	0	+7	+12	+18	+23	+28		+33	+39	+45		+60	

Dimensional class		Tolerance grade										
mm		IT1	IT2	IT3	IT4	IT5	IT6	IT7	IT8	IT9	IT10	IT11
Above		Tolerance										
	Up to	μm										
-	3	0.8	1.2	2	3	4	6	10	14	25	40	60
3	6	1	1.5	2.5	4	5	8	12	18	30	48	75
6	10	1	1.5	2.5	4	6	9	15	22	36	58	90
10	18	1.2	2	3	5	8	11	18	27	43	70	110

Basic size		Lower deviations E_i										
		Tolerance grades (all)										
Above	Till to	A ¹⁾	B ¹⁾	C	CD	D	E	EF	F	FG	G	H
-	3	+270	+140	+60	+34	+20	+14	+10	+6	+4	+2	0
3	6	+270	+140	+70	+46	+30	+20	+14	+10	+6	+4	0
6	10	+280	+150	+80	+56	+40	+25	+18	+13	+8	+5	0
10	14											
14	18	+290	+150	+95		+50	+32		+16		+6	0

○ (a)
Interference

Indicates and explains all the types of sections that can be used in a technical drawing



Section classification

1. In relation with cutting plane

- full section
- offset section
- aligned section

2. In relation with the cutting plane extension

- half sections
- broken out

3. In relation with the cutting plane location

- revolved section
- removed section